

MOUNT KENYA UNIVERSITY

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ADMISSION NUMBER: BSCCS/2024/73613

COURSE: COMPUTER SCIENCE

UNIT NAME: ADVANCE WEB DESIGN AND DEVELOPMENT

LECTURER NAME: MICHAEL NYORO

SEMESTER: YEAR 3 SEM 2

PROJECT TITLE:

TECHNOLOGIES: PHP + MYSQL

GITHUB LINK:

https://github.com/MrsNjenga/my_project.git

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WEEK ONE: LOCAL ENVIRONMENT SETUP

Title: installation and testing of local development environment

N/B: I had already set up the environment earlier

WEEK TWO: WIREFRAMES AND GUI DESIGN

Title: User interface planning and system design

Fig1: login wireframe

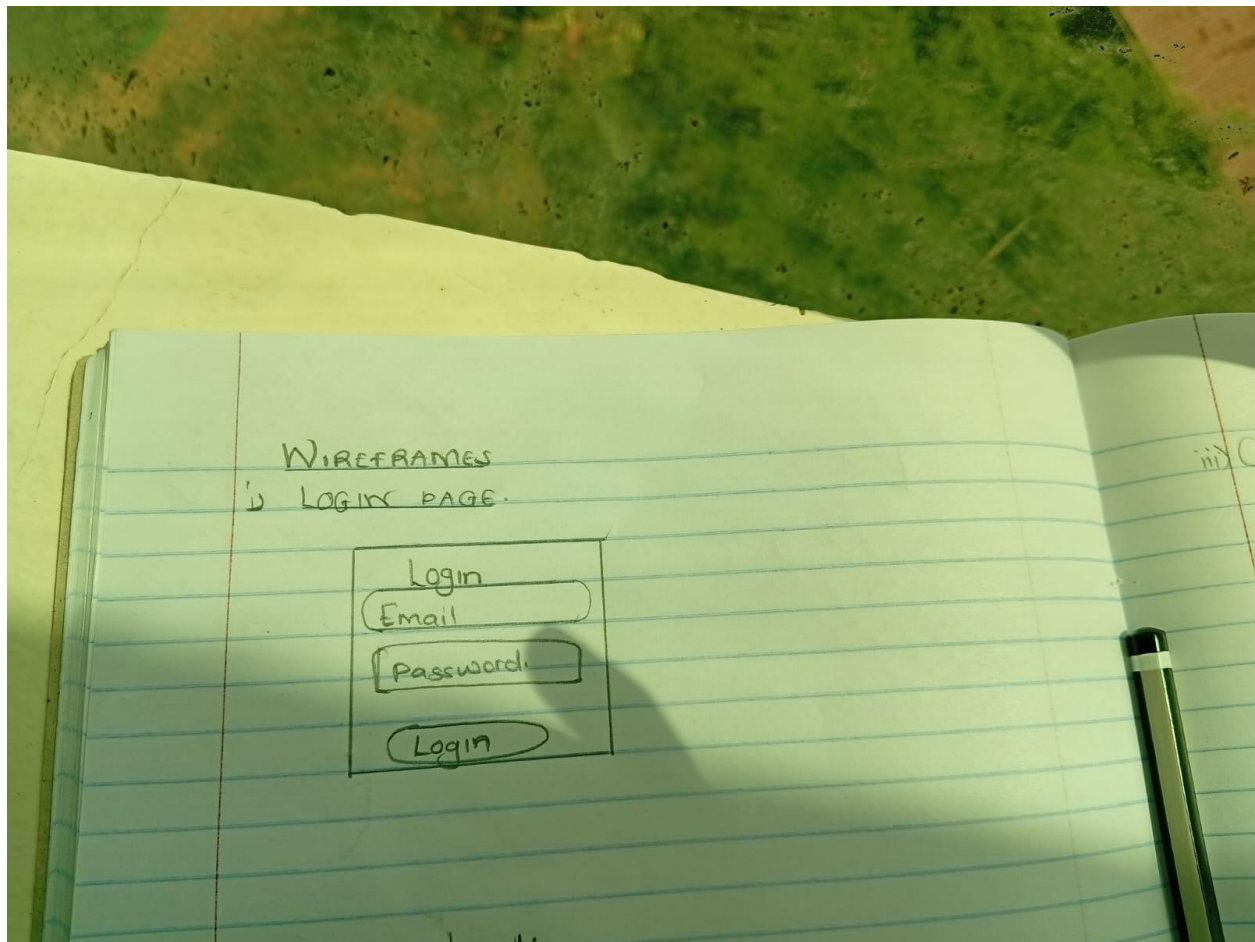


Fig2:Dashboard layout

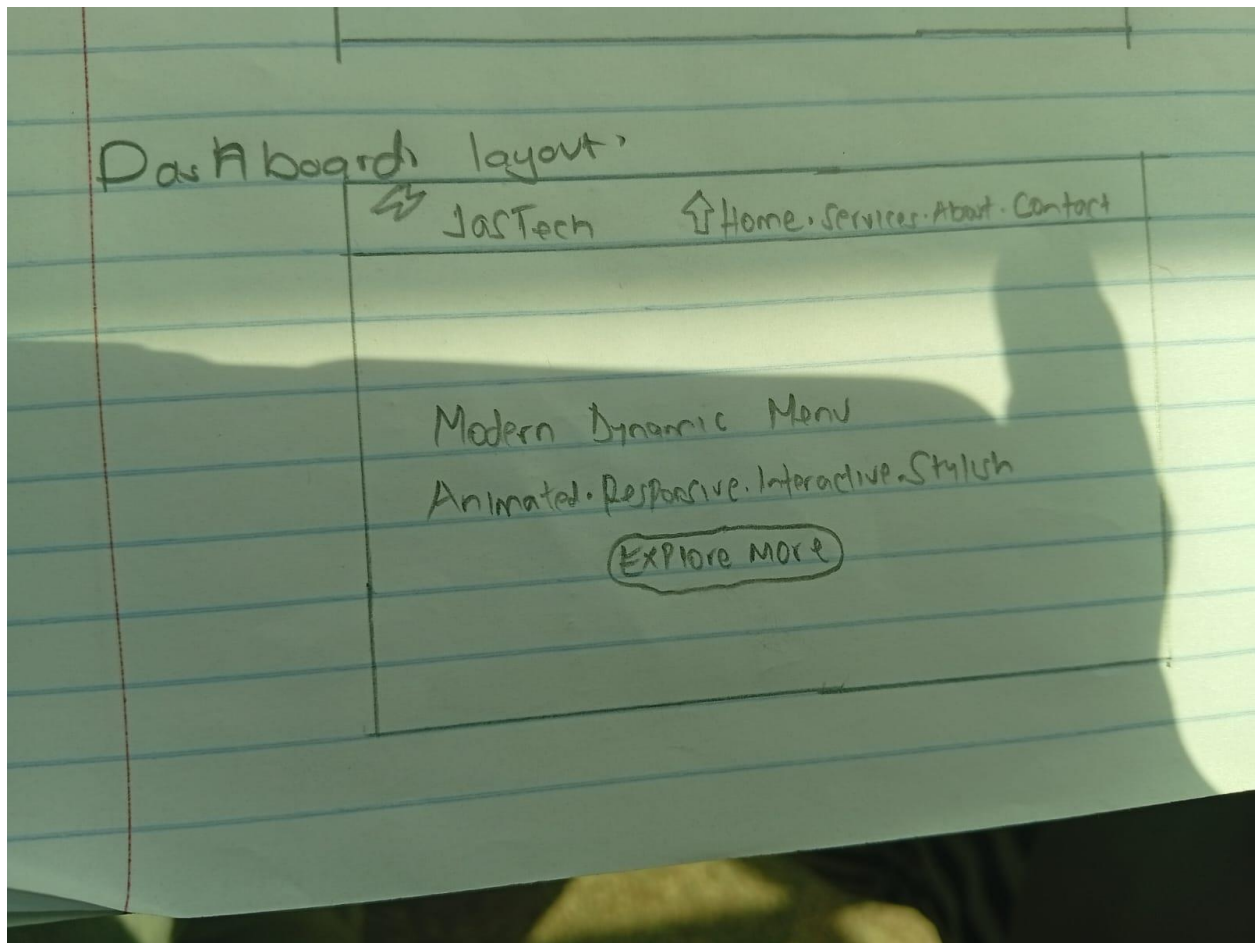
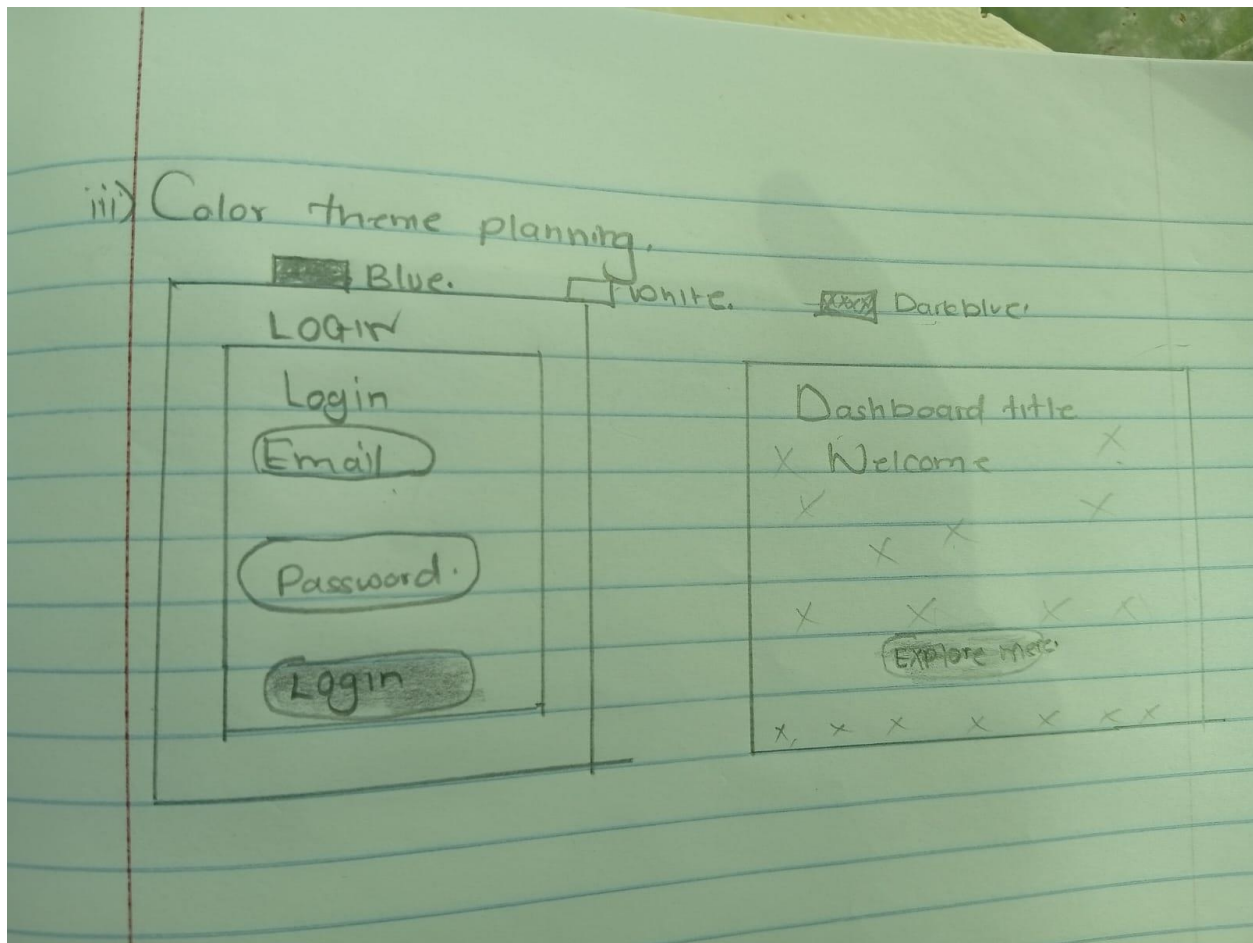


Fig3: Color theme planning

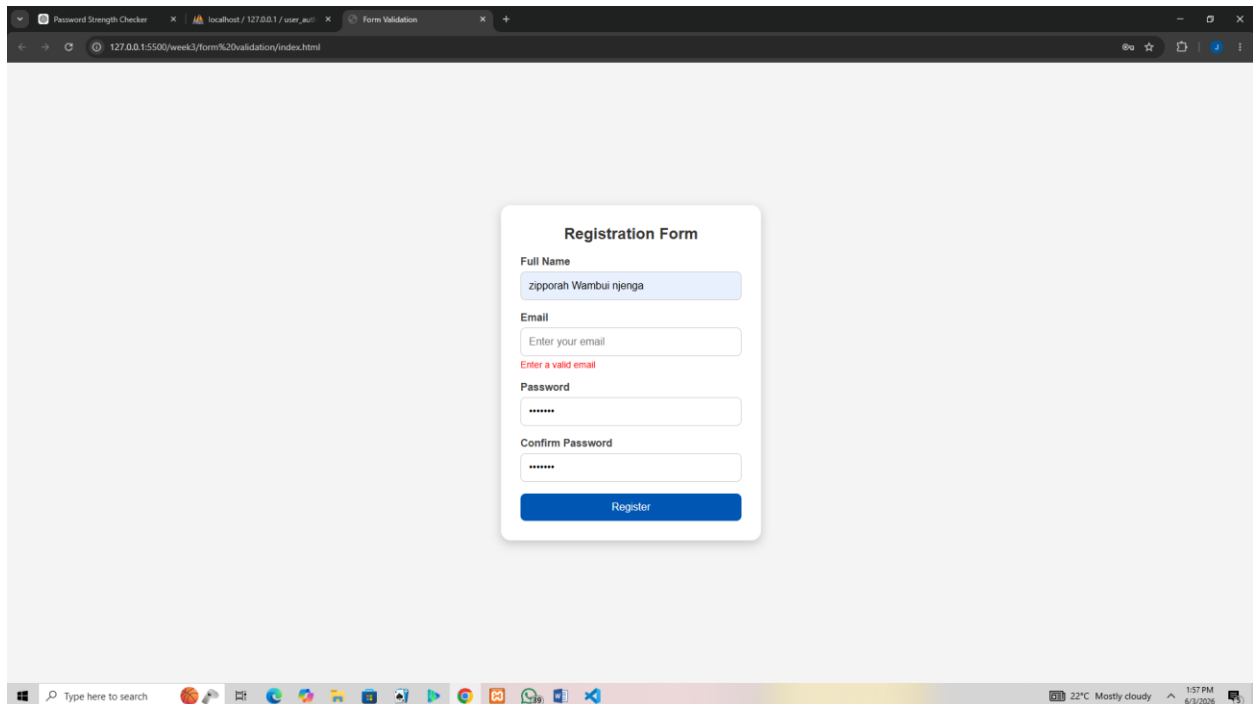


WEEK THREE: JAVASCRIPT AND PHP BASICS

Title: Frontend interaction and Backend

Evidences:

Fig1: Javascript form validation preventing empty submissions



The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5500/week3/form%20validation/index.html". The browser has three tabs: "Password Strength Checker", "localhost / 127.0.0.1 / user_e...", and "Form Validation". The main content area displays a "Registration Form" with the following fields and validation messages:

- Full Name:** A text input field containing "ziporah Wambui njenga".
- Email:** A text input field with the placeholder "Enter your email". Below the field, a red error message reads "Enter a valid email".
- Password:** A text input field with masked characters "*****".
- Confirm Password:** A text input field with masked characters "*****".
- Register:** A blue button at the bottom of the form.

The browser's taskbar at the bottom shows the Windows search bar, various application icons, and system information: "22°C Mostly cloudy", "1:57 PM", and "6/3/2026".

Fig2: Password strength checker with dynamic feedback

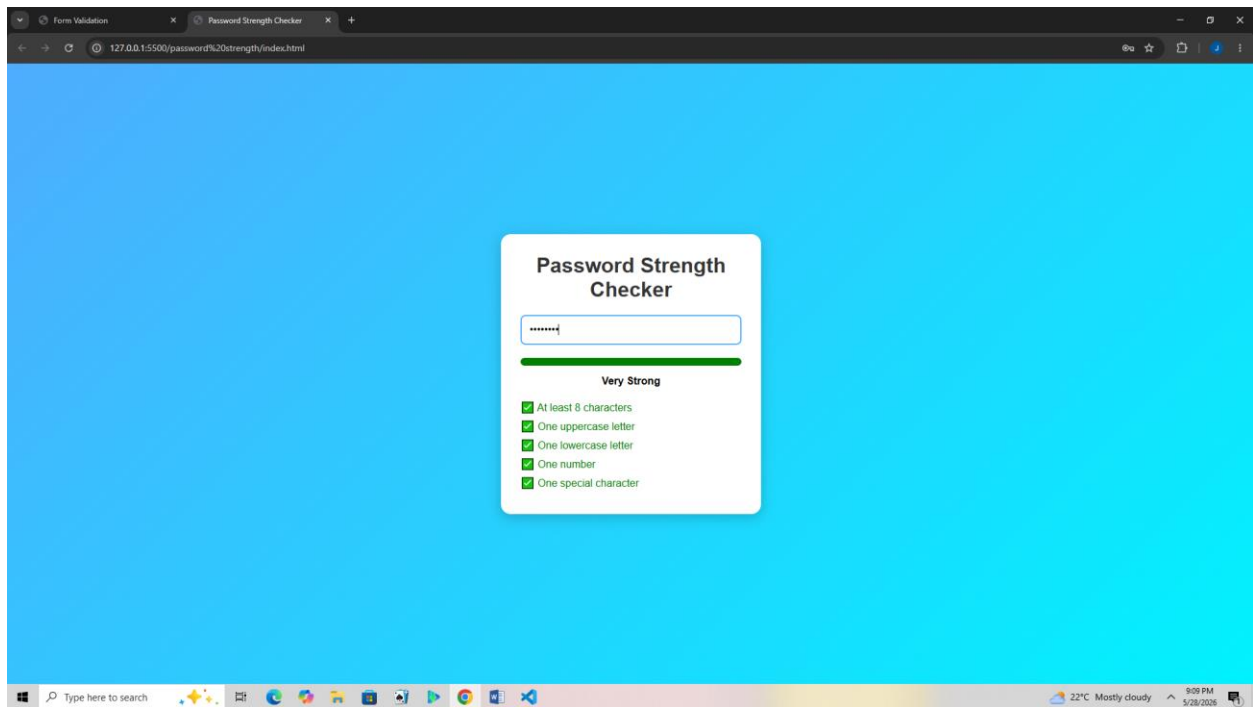


Fig3:Email validation script ensuring correct format

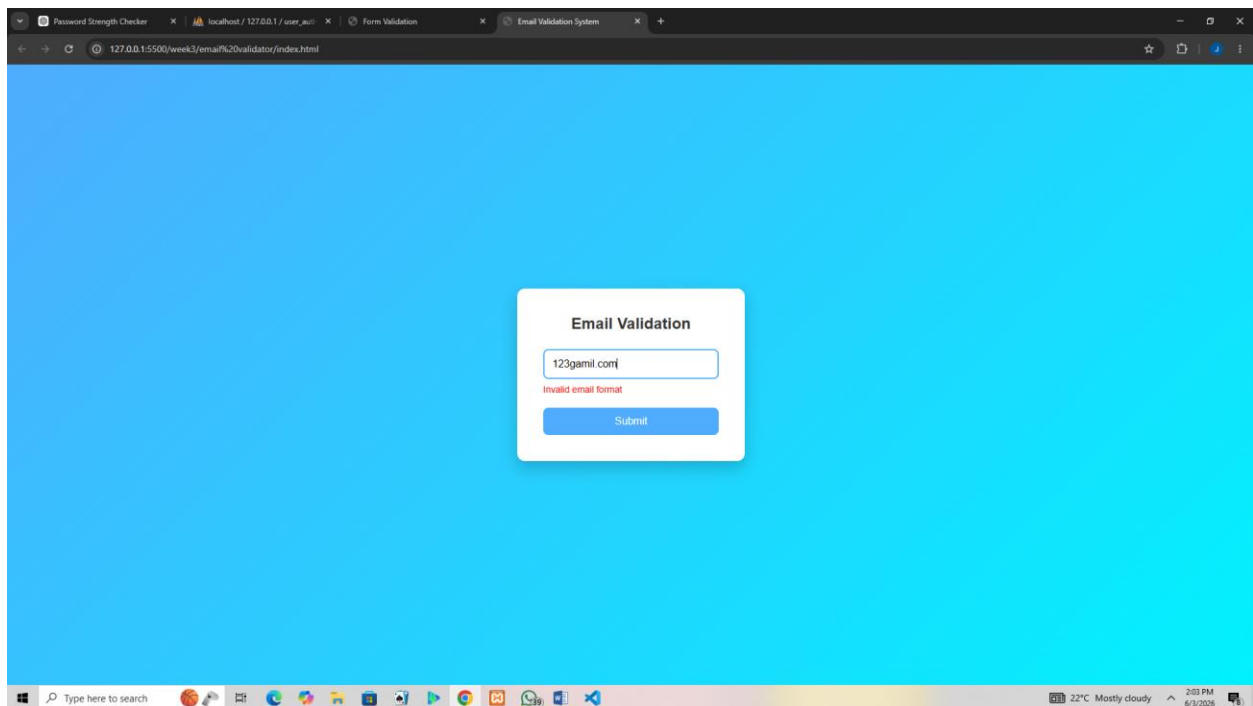


Fig4: login form validation with error message

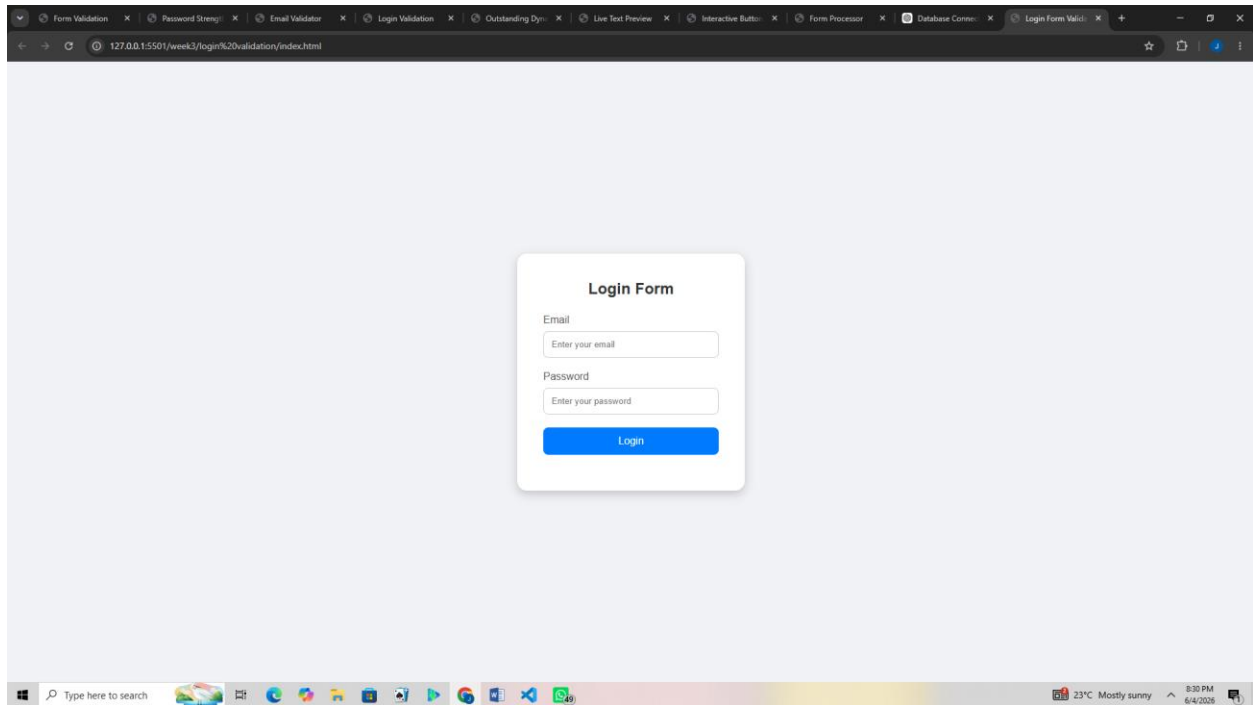


Fig5: Dynamic menu created with javascript

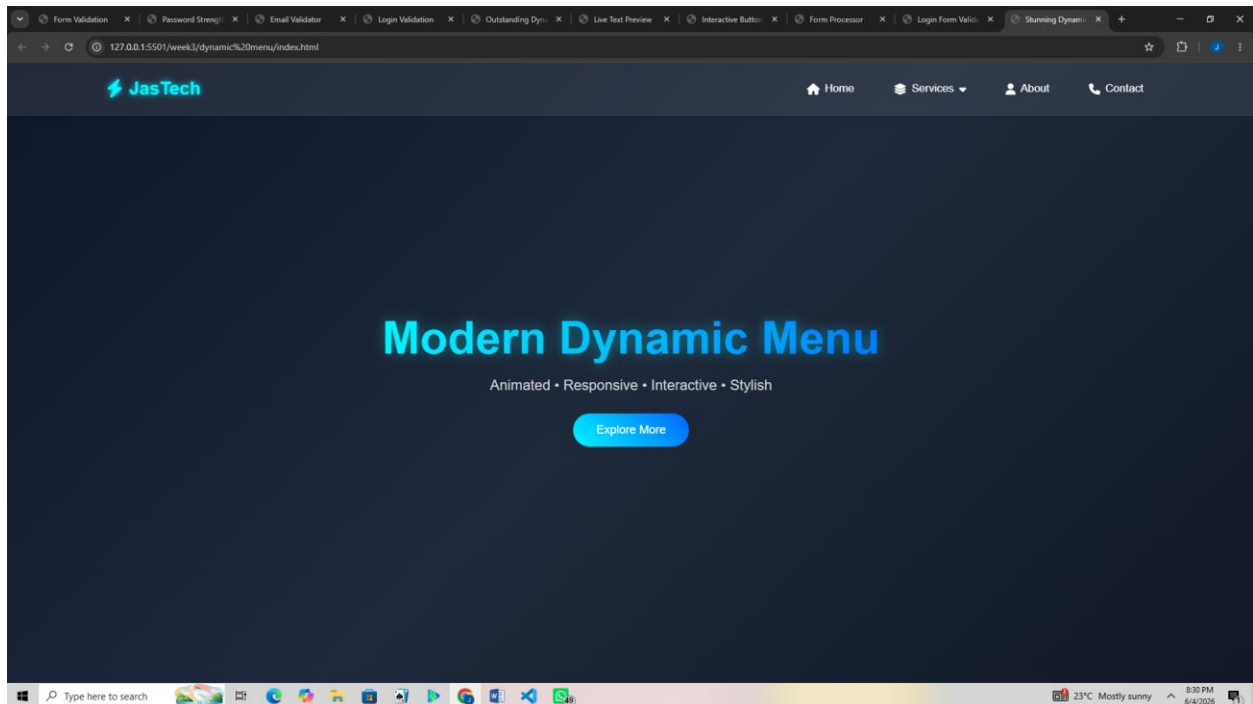


Fig6: live text preview system updating content in real time

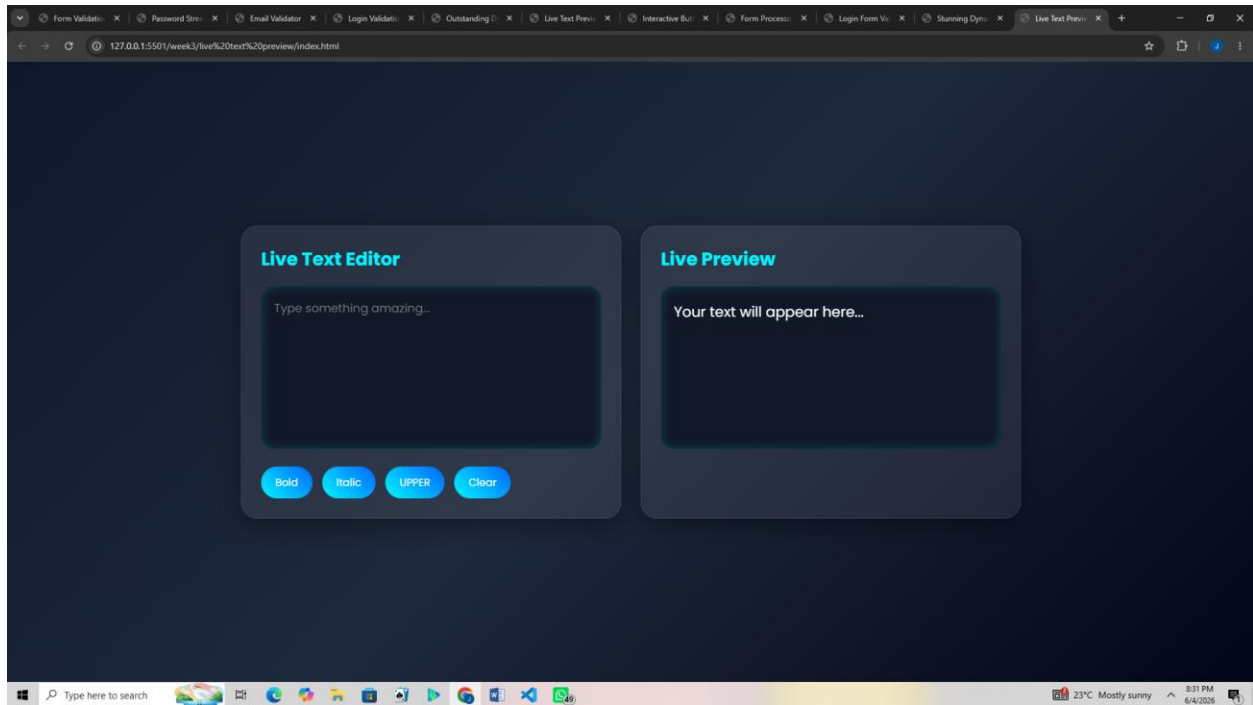
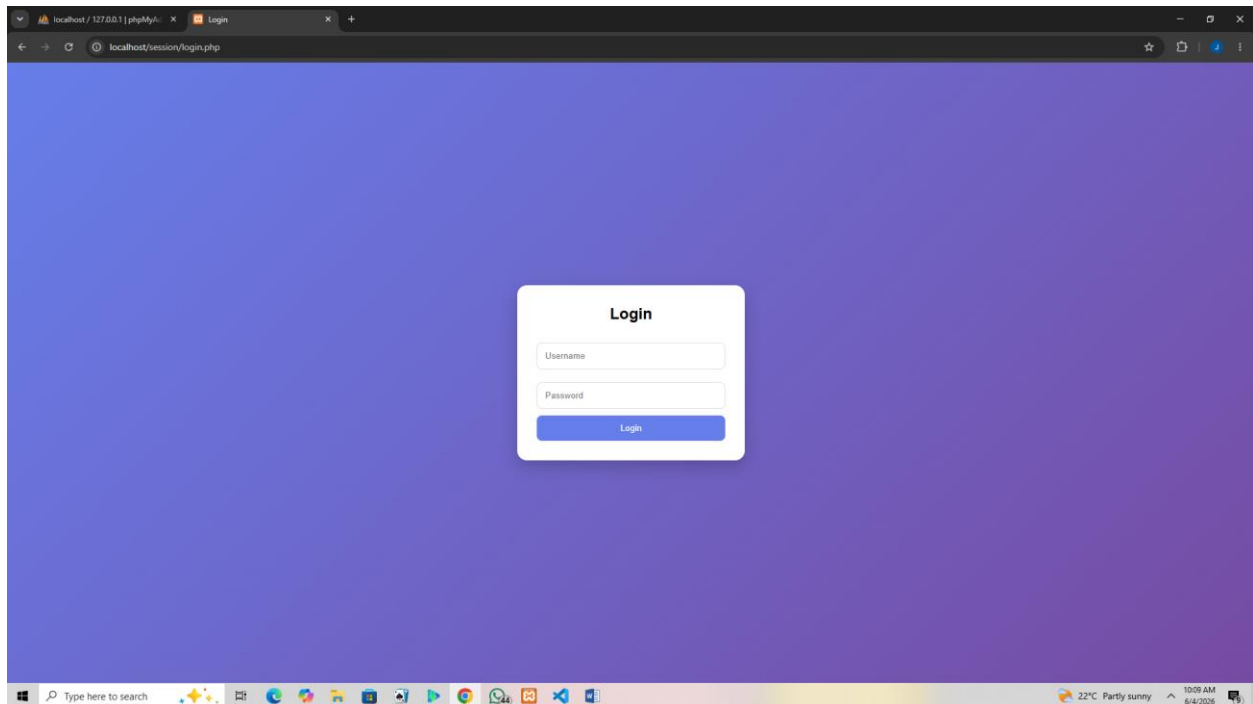


Fig7: index.php file displaying welcome message



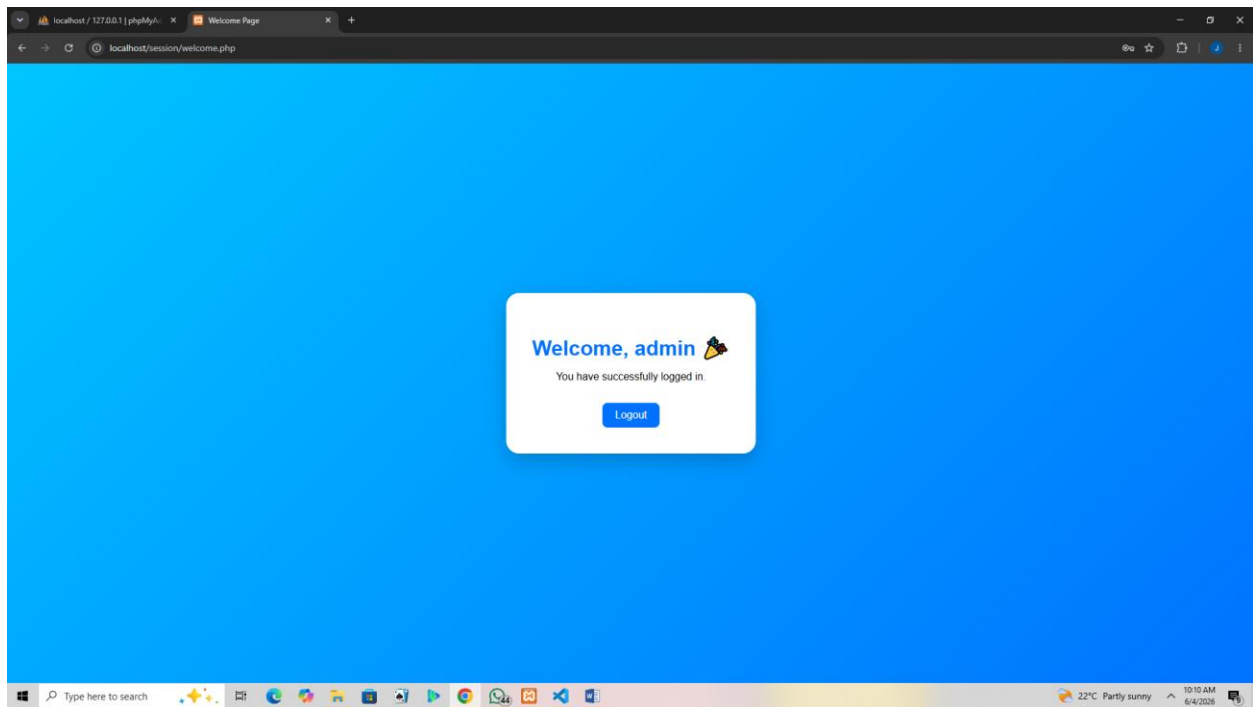


Fig8:interactive buttons responding to script

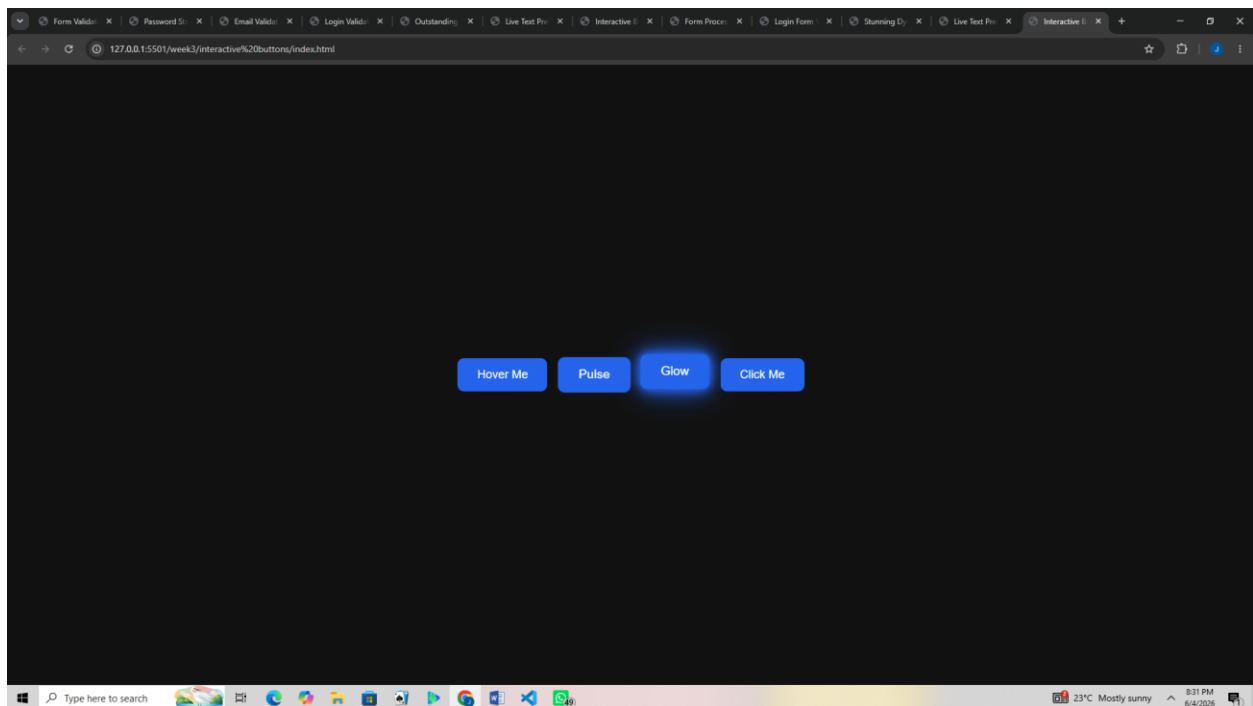
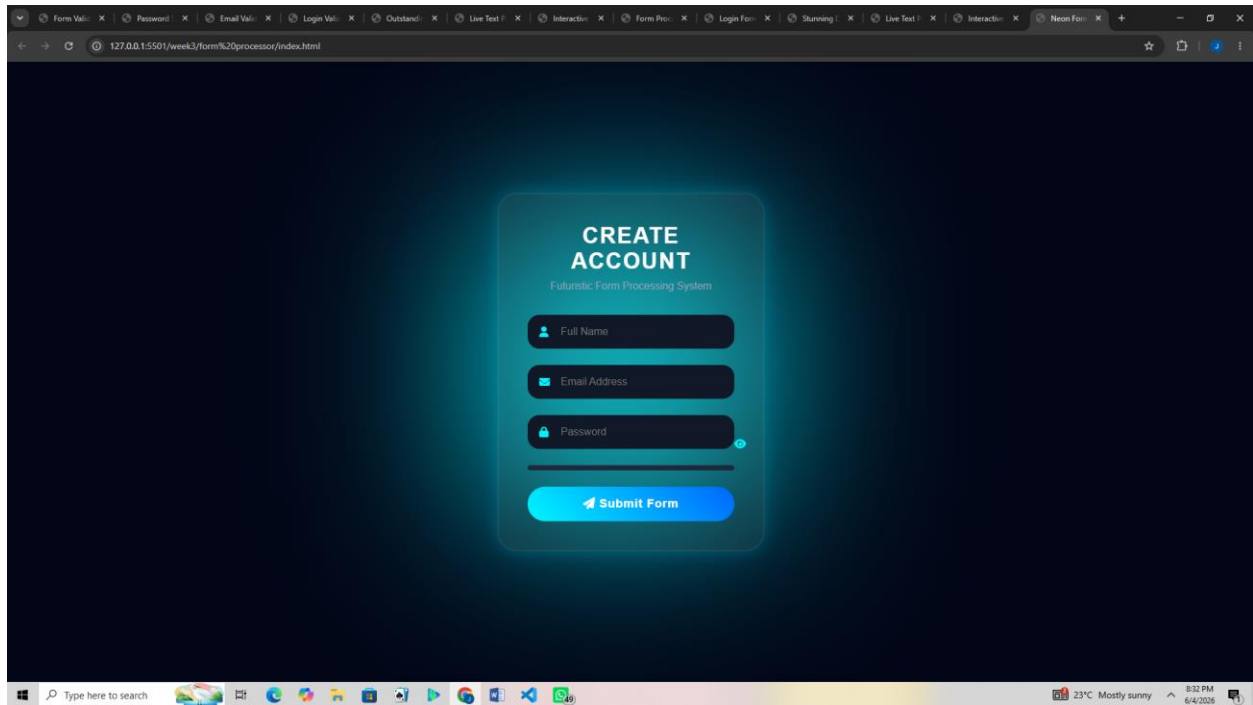


Fig9: simple form processor



REFLECTION

This week I advanced from static pages to interactive systems. JavaScript tasks (Fig 1–7) improved user experience through validation, dynamic menus, and live previews. PHP basics (Fig 8–9) introduced server-side scripting and form handling, while Fig 10 confirmed successful database connectivity. These tasks demonstrated how frontend scripts and backend code integrate to build dynamic applications. The main challenge was debugging validation scripts and PHP syntax errors, which taught me careful attention to detail and improved my problem-solving skills:

WEEK 4:SERVER-SIDE COMPONENTS & BACKEND

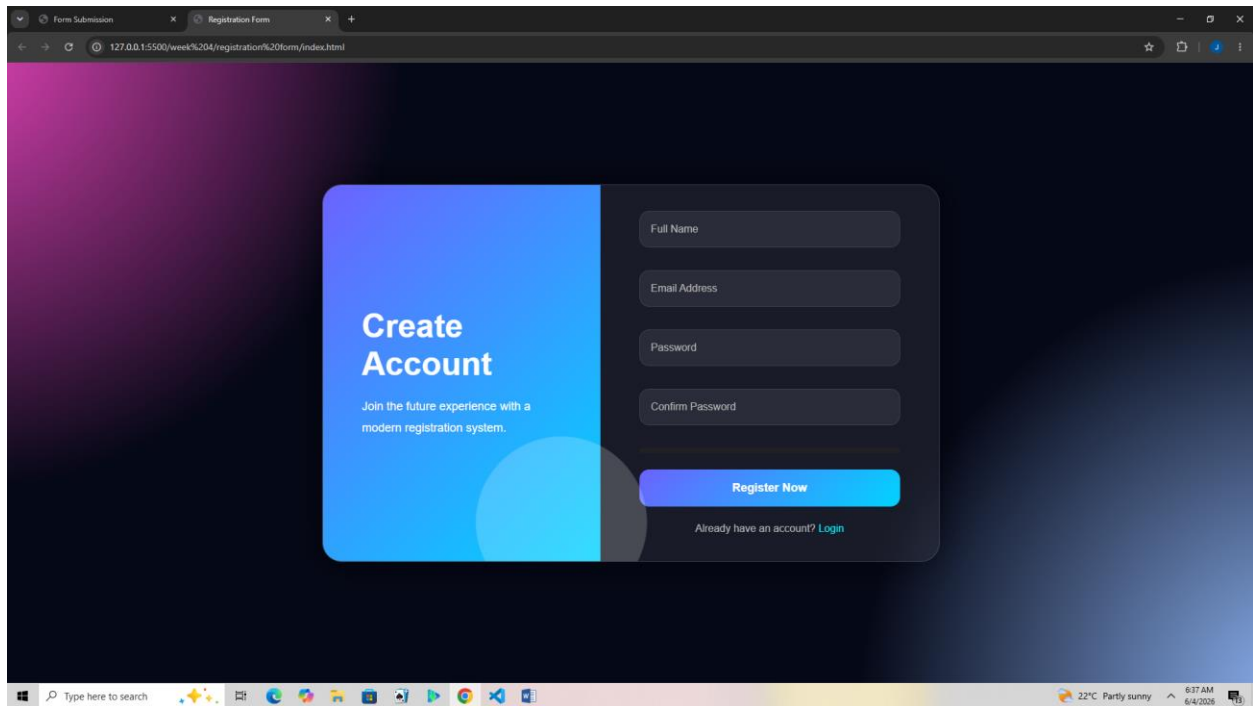
Fig 1: Form submission system displaying dynamic welcome message

The screenshot displays a web browser window with the title "Stunning Form Submission". The address bar shows the URL "127.0.0.1:5500/week%204/form%20submission%20system/index.html". The main content area features a dark blue background with a grid of small, light blue dots. In the center, there is a white, rounded rectangular form with a subtle glow. The form contains the following elements:

- Join The Future**: A heading in bold black text.
- Create your account in style ✨**: A subheading in a smaller, lighter font.
- Full Name**: A text input field with a placeholder label.
- Email Address**: A text input field with a placeholder label.
- Password**: A text input field with a placeholder label.
- Submit**: A button with a blue-to-purple gradient.

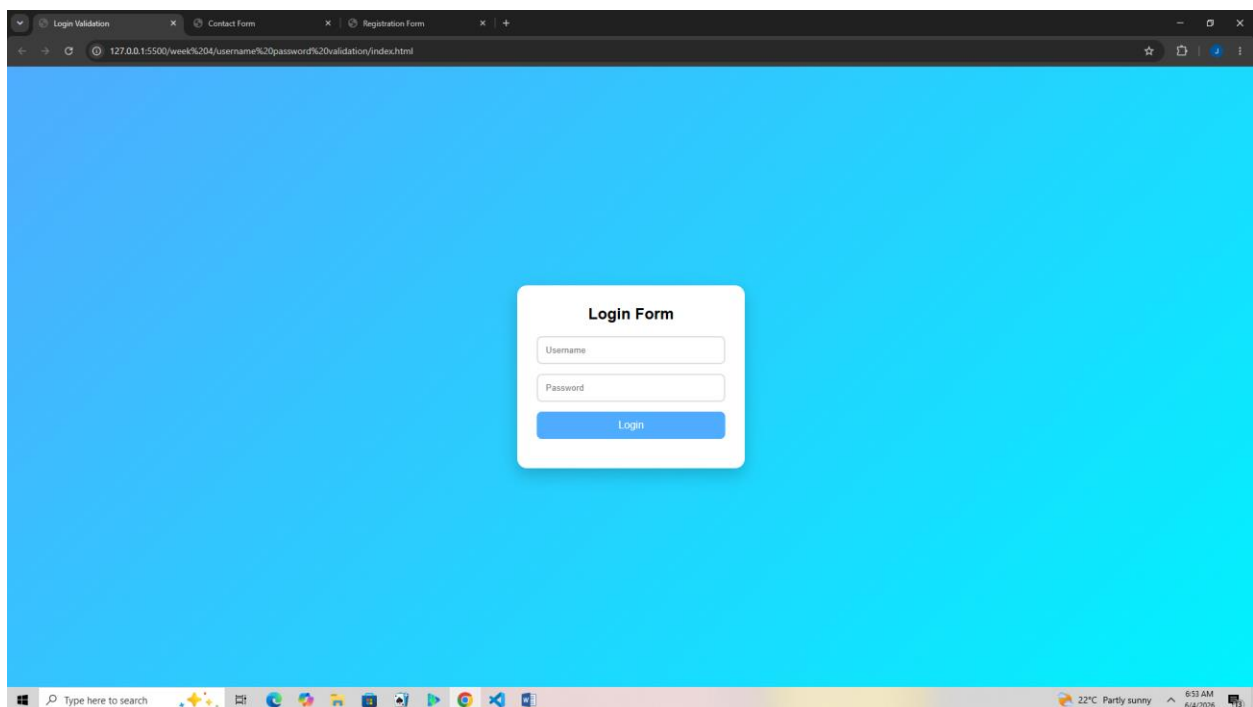
The Windows taskbar is visible at the bottom, showing the search bar, task view button, and several application icons. The system tray on the right indicates a temperature of 22°C, "Partly sunny" weather, and the time 6:36 AM on 6/4/2026.

Fig 3: Registration form integrated with POST method



The screenshot displays a web browser window with the address bar showing the URL `127.0.0.1:5500/week%204/registration%20form/index.html`. The page features a dark blue gradient background. On the left, a blue rounded rectangle contains the text "Create Account" and "Join the future experience with a modern registration system." To the right, a dark gray rounded rectangle contains four input fields labeled "Full Name", "Email Address", "Password", and "Confirm Password". Below these fields is a blue "Register Now" button and a link that says "Already have an account? Login". The Windows taskbar at the bottom shows the search bar, task view button, and several application icons, along with system information indicating 22°C and 6:37 AM on 6/4/2026.

Fig 4: Login form with username/password validation



The screenshot shows a web browser window with the address bar displaying `127.0.0.1:5500/week%204/username%20password%20validation/index.html`. The page has a solid light blue background. In the center, a white rounded rectangle titled "Login Form" contains two input fields labeled "Username" and "Password", followed by a blue "Login" button. The Windows taskbar at the bottom is identical to the one in Fig 3, showing the search bar, task view button, application icons, and system information for 22°C and 6:37 AM on 6/4/2026.

Fig 5: Contact form submission example

Form Submission x Registration Form x Login Form x Contact Form x Contact Form x +

127.0.0.1:5500/week%204/contact%20form/index.html

Contact Us

Send us a message and we'll respond as soon as possible.

Full Name

Email Address

Subject

Your Message

Send Message ✉

Type here to search

22°C Partly sunny 6:44 AM 6/4/2026

Fig 6: Basic login page with conditional access logic

Form Submission x Registration Form x Login Form x Contact Form x Contact Form x Login Form x +

127.0.0.1:5500/week%204/login%20form/index.html

Login

Enter the cyber world with a futuristic neon interface.

Email Address

Password

☒ Remember Me [Forgot Password?](#)

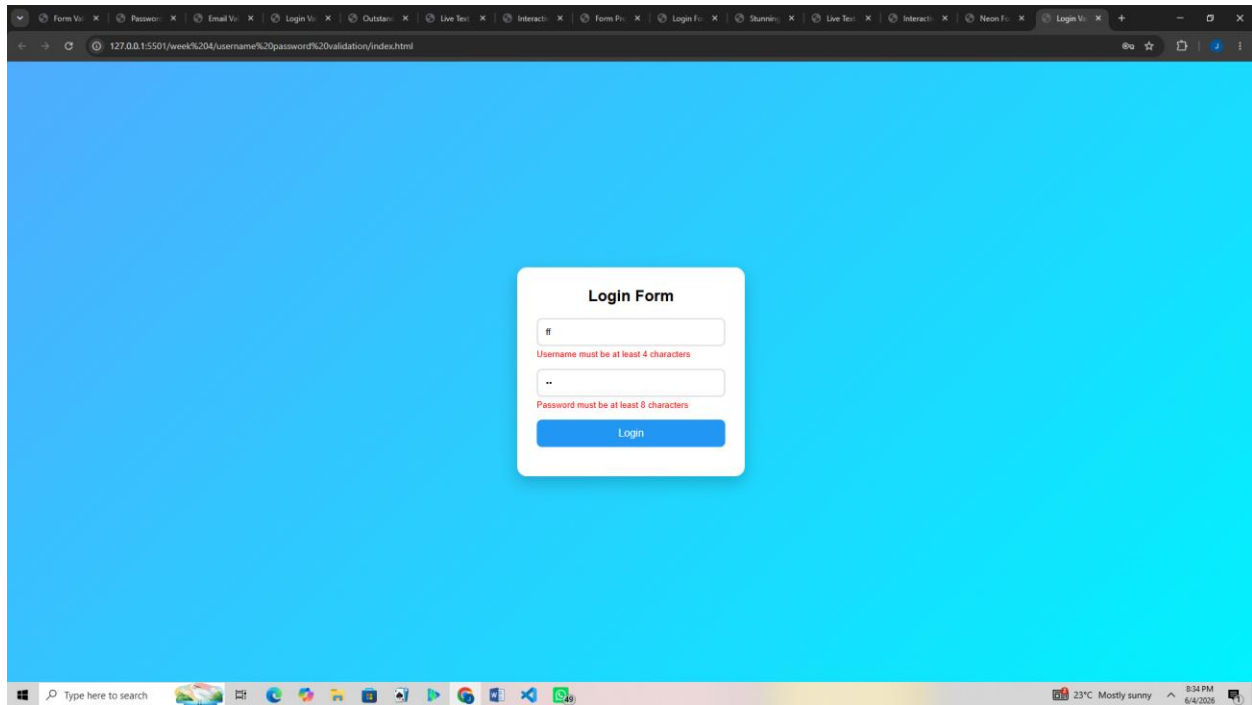
Access System

Don't have an account? [Create Account](#)

Type here to search

22°C Partly sunny 6:45 AM 6/4/2026

Fig 7: Username and password validation script ensuring secure login



REFLECTION

This week I learned how server-side programming processes user requests. Fig 1–3 show PHP handling form submissions and generating dynamic content. Fig 4–6 demonstrate registration, login, and contact forms integrated with backend logic. Fig 7–8 confirm authentication with username/password validation, while Fig 9 shows session handling for personalized user experiences. Organizing files into a backend structure improved workflow. The main challenge was ensuring secure handling of form data, which taught me the importance of using POST methods and validating inputs before granting access

WEEK 5:DATABASE

Fig1: phpMyAdmin showing the database successfully created.

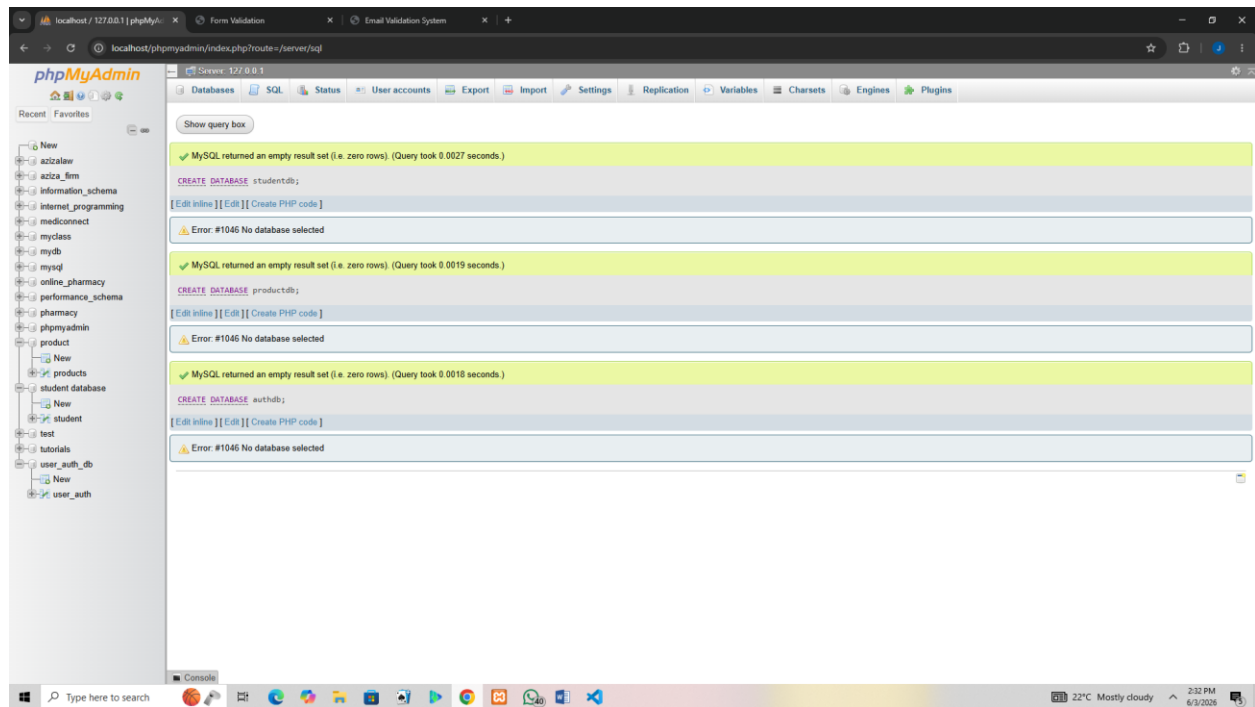


Fig 2: Table creation for users with ID, username, and password fields

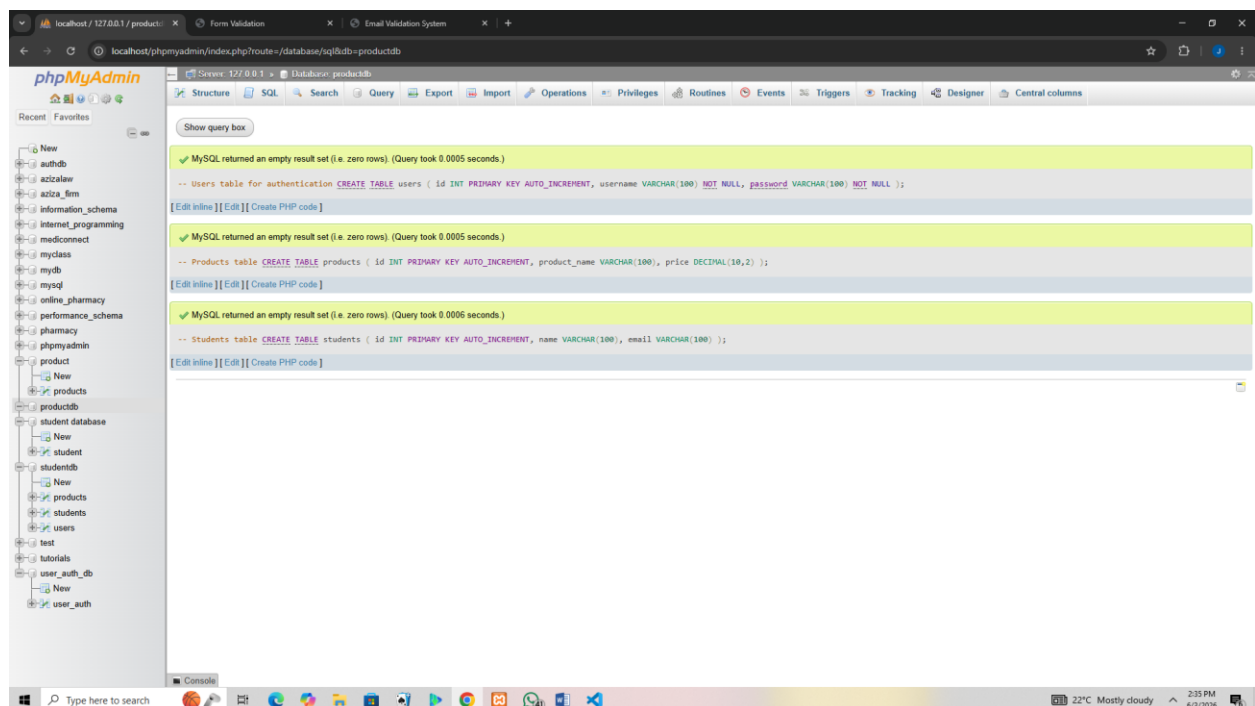


Fig 4: Insert query adding new student record

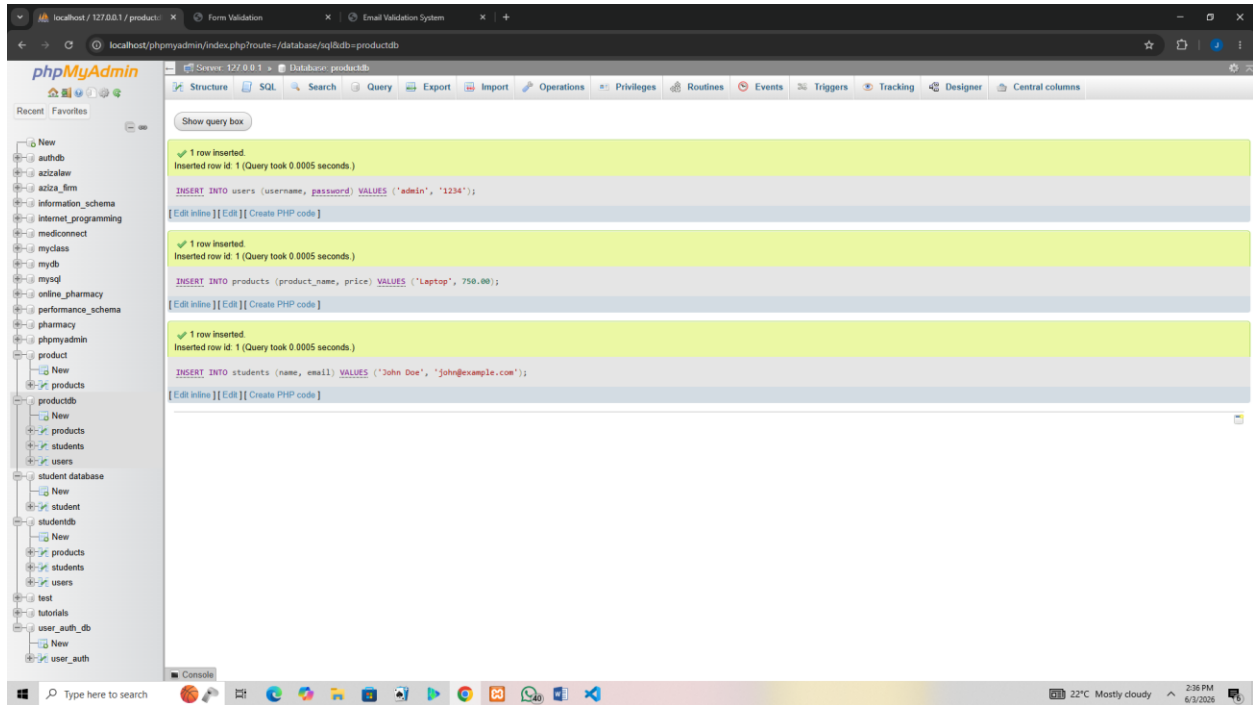
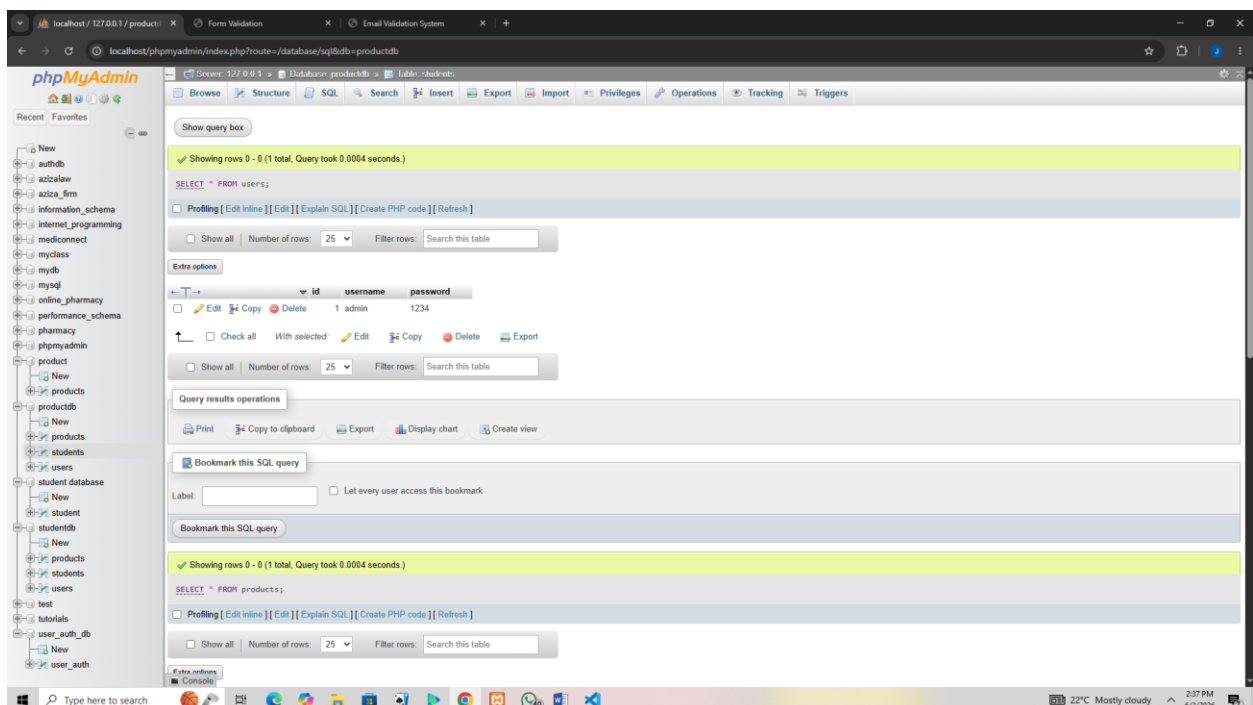


Fig 5: View records query displaying stored data



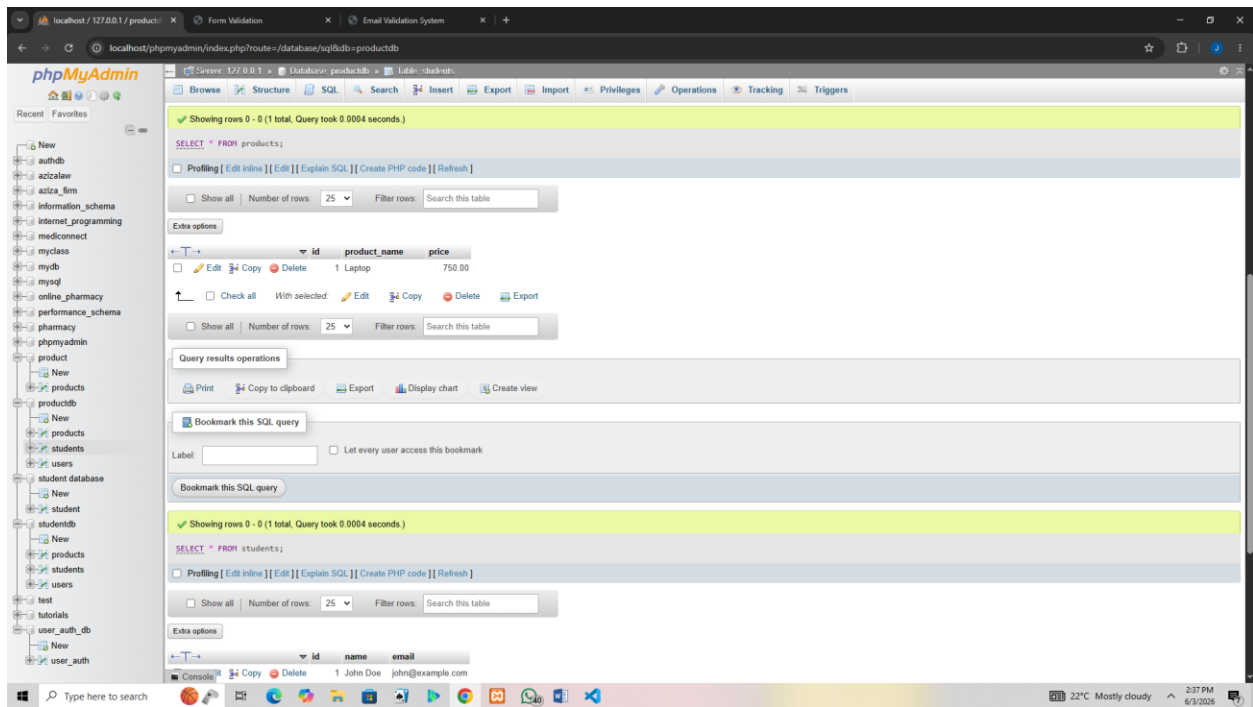


Fig 6: Edit query updating user information

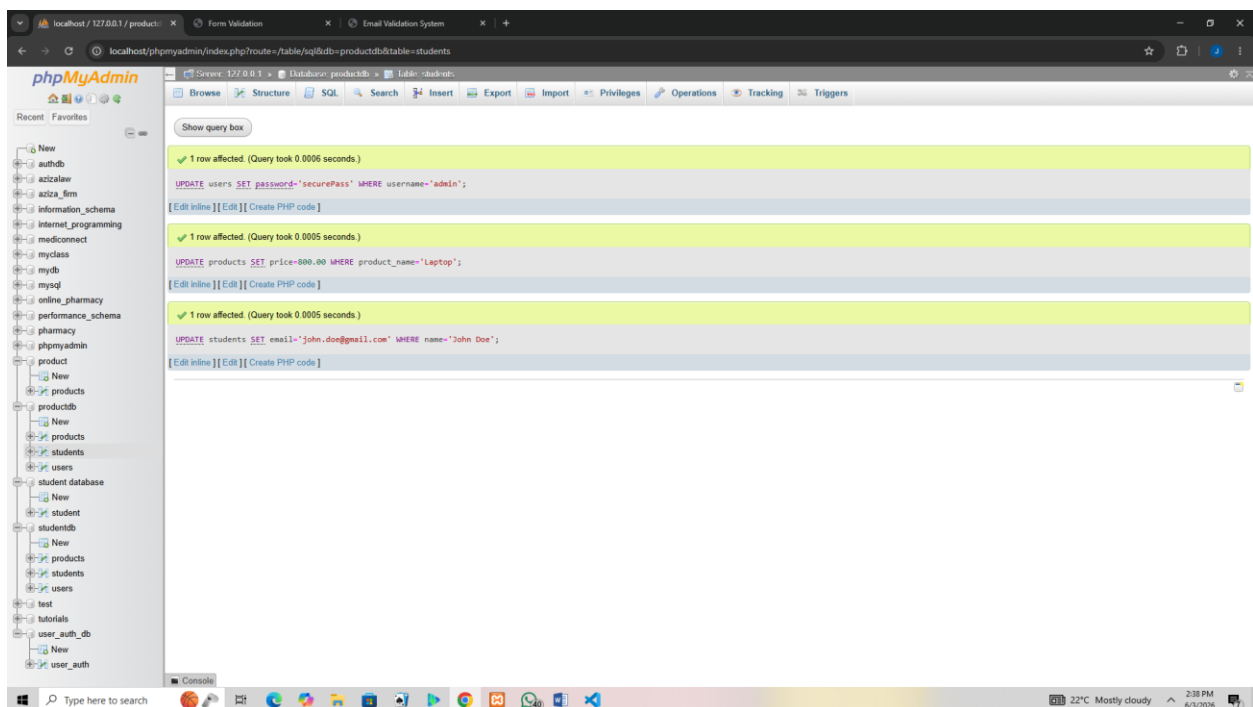


Fig 8: Delete query removing a record successfully(codes)

```
DELETE FROM users WHERE username='admin';
```

```
DELETE FROM products WHERE product_name='Laptop';
```

```
DELETE FROM students WHERE name='John Doe';
```

REFLECTION

This week I learned how to build database-driven systems. Fig 1–3 show the creation of student, product, and authentication databases. Fig 4 demonstrates table setup for user credentials. CRUD operations (Fig 5–8) allowed me to add, view, edit, and delete records, proving full database functionality. These tasks taught me how backend systems manage persistent data and connect to frontend forms. The main challenge was ensuring queries executed correctly, which improved my understanding of SQL syntax and database integration with PHP.

